



Module	SE101.3 -OOP with Java – 25.3
Lab sheet Number	11
Areas covered	Revision

Part A – Multiple Choice Questions

Question 1

Which of the following declarations is **NOT** valid in Java?

a)

```
abstract class A {  
    abstract void display();  
}
```

b)

```
abstract class B {  
    static void show(){}  
}
```

c)

```
abstract class C {  
    static abstract void print();  
}
```

d)

```
abstract class D {  
    void run(){}  
}
```

Question 2

Consider the following code.

```
abstract class Vehicle {  
    Vehicle() {  
        System.out.println("Vehicle");  
    }  
}  
  
class Car extends Vehicle {  
    Car() {  
        System.out.println("Car");  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Car car = new Car();  
    }  
}
```

What is the output?

- a) Car
Vehicle
- b) Vehicle
Car
- c) Vehicle
- d) Compilation Error

Question 3

Which statement about constructors in an abstract class is correct?

- a) Constructors cannot be declared inside abstract classes.
- b) Constructors execute only if the abstract class is instantiated directly.
- c) Constructors are executed when a subclass object is created.
- d) Constructors in abstract classes must be abstract.

Question 4

Consider the following program.

```
abstract class A {  
    {  
        System.out.println("A");  
    }  
}  
  
class B extends A {  
    {  
        System.out.println("B");  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        new B();  
    }  
}
```

What is the output?

- a) B
A
- b) A
B
- c) A
- d) Compilation Error

Question 5

Which keyword **cannot** be used with an abstract method?

- a) public
- b) protected
- c) final
- d) default

Question 6

Which of the following statements is **correct** about an abstract class in Java?

- a) An abstract class can be instantiated directly.
- b) An abstract class cannot contain constructors.
- c) An abstract class can contain both abstract and concrete methods.
- d) Every method in an abstract class must be abstract.

Question 7

Consider the following code.

```
abstract class Shape{
    abstract void draw();
}

class Circle extends Shape{
    void draw(){
        System.out.println("Circle");
    }
}

public class Main{
    public static void main(String[] args){
        Shape s = new Circle();
        s.draw();
    }
}
```

Which concept allows Circle's method to execute?

- a) Constructor Overloading
- b) Runtime Polymorphism
- c) Encapsulation
- d) Composition

Question 8

Which statement about abstract classes is **FALSE**?

- a) An abstract class may contain constructors.
- b) An abstract class may contain concrete methods.
- c) An abstract class can be declared as a local inner class.
- d) Every abstract class must contain at least one abstract method.

Question 9

Consider the following code.

```
abstract class Employee{
    abstract void work();
}

class Developer extends Employee{
    @Override
    void work(){
        System.out.println("Coding");
    }
}

public class Main{
    public static void main(String[] args){
        Employee e = new Developer();
        e.work();
    }
}
```

What is printed?

- a) Employee
- b) Developer
- c) Coding
- d) Compilation Error

Question 10

Which statement best explains **why abstract methods cannot be declared as final**?

- a) Final methods are executed faster.
- b) Final methods cannot be inherited.
- c) Final methods cannot be overridden, whereas abstract methods require overriding.
- d) Abstract methods cannot contain parameters.

Part B – Coding Questions

Question 1

A university is developing a course registration system. Each student has a student ID, name, and current credit count.

A student can register for a new course by entering the number of credits offered by the course.

The system must satisfy the following requirements:

- Store student details using appropriate access modifiers.
- Prevent registration if the total registered credits exceed 24 credits.
- Prevent registration if the entered credit value is zero or negative.
- If either situation occurs, display an appropriate error message and allow the program to continue.
- Display the student's updated credit count after a successful registration.

Sample Output:

Enter Student ID : S001

Enter Student Name : John

Current Credits : 18

Enter Course Credits : 8

Registration Failed : Credit limit exceeded.

Question 2

A smart parking system manages different categories of vehicles entering a parking area.

Each vehicle has:

- Registration Number
- Owner Name

The parking fee is calculated differently depending on the vehicle category.

Develop a Java application that allows the administrator to:

- Register different vehicle types.
- Display vehicle information.
- Calculate and display the parking fee based on the number of parking hours.
- Demonstrate the system using **at least three different vehicle objects**.
- Display the parking fee for each vehicle without using multiple if-else or switch statements to determine the vehicle category.

Question 3

A university employs different categories of staff such as **Academic Staff, Administrative Staff, and Visiting Lecturers**.

Each employee has:

- Employee ID
- Employee Name
- Department

The salary calculation differs depending on the employee category.

Develop a Java application that allows the HR department to:

- Store employee information.
- Calculate the monthly salary.
- Display employee details and salary.
- Demonstrate the payroll system using **at least three different employee objects**.
- Generate the payroll report without using multiple if-else or switch statements to determine the employee category.

Your design should make it easy to introduce new employee categories in the future with minimal modifications to the existing code.